# Theory of Computation

## CO-3 Assignment

PI - 4

COS - 3

AD - 2

TM - 4

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Meetma

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Design a pushdown Automata to recognize the language

$$L = \{ a^n b^p c^p d^{(n/2)} \mid p \geq 1, n > 1 \}$$

Solution

Let the PDA be

$$M = \{ Q, \Sigma, \Gamma, \delta, q_0, z_0, F \}$$

$$\Sigma = \{ a, b, c, d \}$$

$$\Gamma = \{ A, B, z_0 \}$$

$\delta(q_0, a, z_0) = (q_1, z_0)$

$\delta(q_1, a, z_0) = (q_0, Az_0)$

$\delta(q_0, a, A) = (q_1, A)$

$\delta(q_1, a, A) = (q_0, AA)$

$\delta(q_0, b, A) = (q_2, BA)$

$\delta(q_2, b, B) = (q_2, BB)$

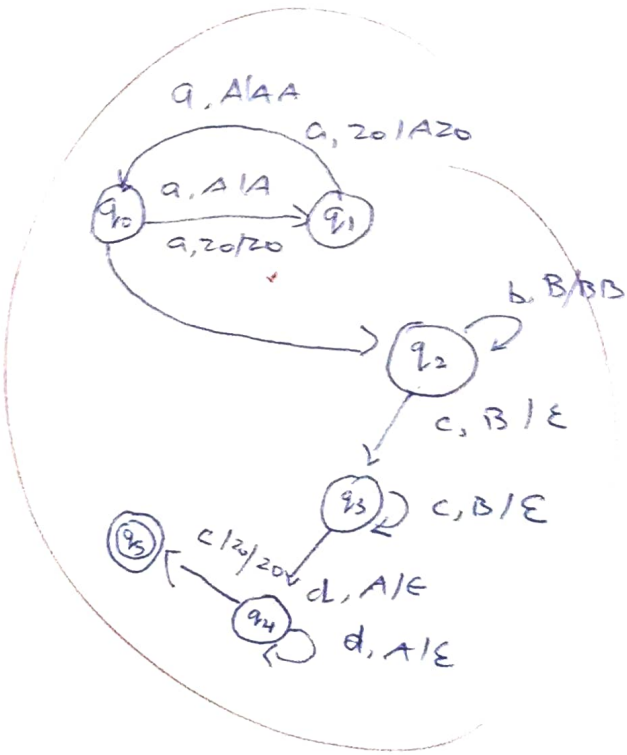
$\delta(q_2, c, B) = (q_3, \epsilon)$

$\delta(q_3, c, B) = (q_3, \epsilon)$

$\delta(q_3, d, A) = (q_4, \epsilon)$

$\delta(q_4, d, A) = (q_4, \epsilon)$

$\delta(q_4, \epsilon, z_0) = (q_5, z_0)$




Execution Trace

$\rightarrow q_0 aaaabbccdd - z_0$

$\rightarrow a q_1 aaabbccdd - z_0$

$\rightarrow aa q_0 aabbccdd - Az_0$

$\rightarrow aaa q_1 abbccdd - Az_0$

$\rightarrow aaaa q_0 bbccdd - AAz_0$

$\rightarrow aaaab q_2 bcdd - BAAz_0$

$\rightarrow aaaabb q_2 cdd - BBAAz_0$

$\rightarrow aaaabbc q_3 dd - Az_0$

$\rightarrow aaaabbcdq_4 d - z_0$

$\rightarrow aaaabbccdq_5 - z_0$

How PDA is constructed into CFG ? (Answer the following PDA to CFG ?

$P = \{\{P, q, r, s\}, \{a, b\}, \{a, z\}, \{P, z, \phi\}, \delta(P, q, z)$

$(P, a, z)$

$\delta(P, a, a) = (P, aa), \delta(P, b, a) = (q, a),$

$\delta(q, b, a) = (r, \varepsilon)$ $\delta(r, b, a) = (a, a), \delta(r, q, z) =$

$(s, z)$

Solution:

Let the CFG be

$G = (V, \Sigma, R, S)$

Let $q_0 = P$, $z_0 = z$

There are totally 32 possible variables for the stack Symbol = stack Alphabets combined

Step 1: To create ~~start~~ start production.

$S \rightarrow [P, z, P]$

$S \rightarrow [P, z, q]$

$S \rightarrow [P, z, r]$

$S \rightarrow [P, z, s]$

$[P, a, P] \rightarrow a [P, q, P] [P, q, P] / a [P, z, q] [q, a, b]$

$a [P, q, r] [r, a, P] / a [P, a, s] [s, a, b]$

$[P, a, q] \Rightarrow a [P, q, P] [P, a, q] / a [P, q, q] [q, a, q]$

$a [P, q, r] [r, a, q], a [P, q, s] [s, a, q]$

$\delta(p, b, a) = (q, a)$

$\delta(q, b, a) = (r, \varepsilon)$

$[q, a, r] \rightarrow b$

$\delta(r, b, a) = (q, a)$

$[r, a, z] \rightarrow b[q, a, z]$

$\delta(r, \varepsilon, z) = (s, z)$

$[r, z, s] = [s, z, s] \rightarrow \varepsilon$

After removing variables ~~~~ that are non-generating

The simplified Grammar is:

$S \rightarrow [p, z, s]$

$[p, z, s] \rightarrow a\,[p, q, r]\,[r, z, s]$

$[p, q, r] \rightarrow a\,[p, a, r]\,[r, q, r]$

$[p, a, r] \rightarrow b\,[q, a, r]$

$[r, a, r] \rightarrow b\,[q, a, r]$

$[q, a, r] \rightarrow b$

$[r, z, s] \rightarrow \varepsilon$

# TOC

## CO-4 Assignment

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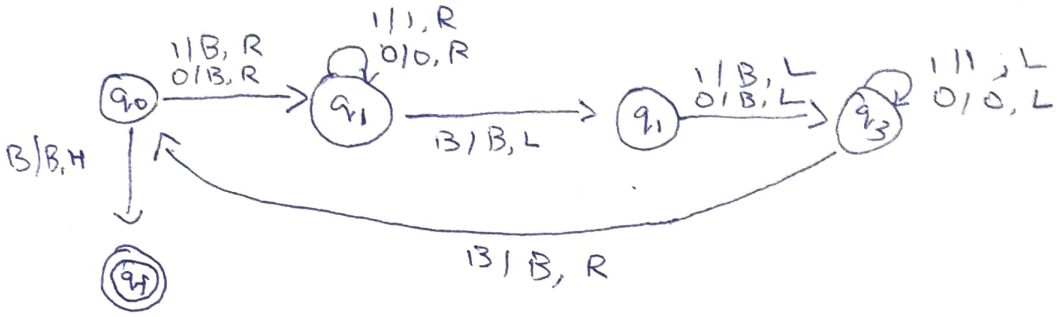
sign a T.M to accept Language L $\{ww^R / w^R$ is reverse of w

race the ID's of TM for the string 010010

$L = \{w\, w^R / w^R$ is Reverse of $w\}$

$L = \{0110,\ 011110,\ 010010,\ 100001, \ldots\}$

uring Machine:




For the string 010010

| B | 0 | 1 | 0 | 0 | 1 | 0 | B |
|---|---|---|---|---|---|---|---|

$\vdash q_0\ 010010B$

$\vdash B\ q_1\ 10010B$

$\vdash B\ 10010\ q_1\ B$

$\vdash B\ 1001\ q_3\ B\ B$

$\vdash q_0 B\ 1001\ B\ B$

$\vdash B B\ q_2\ 001 B\ B$

$\vdash B B 001\ q_2\ B B$

$\vdash B B 00\ q_3\ B B B$

$\vdash q_0 B B 00 B B$

$\vdash B B B\ q_1\ 0 B B B$

$\vdash B B B\ q_3\ B B B B$

$\vdash q_0\ B B B B B B B$

$\vdash q_1\ B B B B B B B$

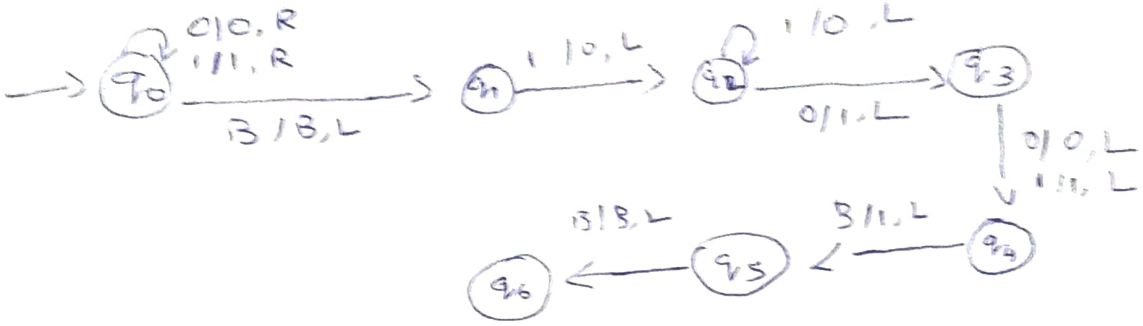
$\therefore$ 010010 is accepted

2. Design a TM that takes a Binary number as input and will increment the number by one

100 + 1 → 101

101 + 1 → 110

Turing Machine




Let input = 1011

| B | 1 | 0 | 1 | 1 | B |
|---|---|---|---|---|---|

$\vdash q_0 1011$

$\vdash 1 q_0 011$

$\vdash 10 q_1 11$

$\vdash 101 q_1 B$

$\vdash 10 q_2 1$

$\vdash 1 q_2 00$

$\vdash q_2 100$

$\vdash q_f 1100$

| B | 1 | 1 | 0 | 0 | B |
|---|---|---|---|---|---|

∴ The output string

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